

CMSC828N: Logical Aspects of AI - Homework 2

Peratham Wiriyathamabhum

1 Question 1

Suppose P is a logic program. An Herbrand interpretation I is a “strong model” for P ($I \models_{strong} P$) iff $I \models P$ and for all ground atoms A in I , it is the case that there exists a rule in P having a ground instance of the form $A \leftarrow B_1, \dots, B_n$ such that $\{B_1, \dots, B_n\} \subseteq I$.

Is the following statement correct? *I is a strong model of P iff $T_p(I) = I$.* If it is correct, please provide a proof - otherwise please provide a counter example.

Answer

For $(\rightarrow)$: ($I \models_{strong} P$) $\rightarrow$ ($T_p(I) = I$).

- Suppose $I \models P$ and $A \in T_p(I)$.
- Then there is a rule in P of the form $A \leftarrow B_1, \dots, B_n$ such that $\{B_1, \dots, B_n\} \subseteq I$.
- As $I \models P$ and $\{B_1, \dots, B_n\} \subseteq I$, $A \in I$.
- This means $T_p(I) \subseteq I$.
- As $I \models_{strong} P$, repeat the previous 2 steps to $\forall A \in I$.
- There exists a rule in P of the form $A \leftarrow B_1, \dots, B_n$ such that $\{B_1, \dots, B_n\} \subseteq I$ for every A .
- Then $I \subseteq T_p(I)$ because $A \in T_p(I)$ and $I = \cup_i \{A_i\}$, hence $T_p(I) = I$.

For $(\leftarrow)$: ($T_p(I) = I$) $\rightarrow$ ($I \models_{strong} P$).

- Suppose $T_p(I) = I$ and $A \leftarrow B_1, \dots, B_n$ is a ground instance r of a rule in P such that $\{B_1, \dots, B_n\} \subseteq I$.
- We need to show that $A \in I$.
- As $\{B_1, \dots, B_n\} \subseteq I$, $A \in T_p(I)$.

- As $T_p(I) = I$, $A \in I$.
- This means I satisfies r so $I \models P$.
- As $I \models_{strong} P$, we need to also show that for $\cup_i \{A_i\} = I$ there exist r for each A_i .
- Since $T_p(I) = I$, $A \in lfp(T_p)$. We show that $P \models A$.
- As $lfp(T_p) = lpre(T_p)$, $A \in$ each pre-fixpoint of T_p .
- Since models of P are pre-fixpoint of T_p , $P \models A$.
- In other words, there exists a rule r in P of the form $A \leftarrow B_1, \dots, B_n$ such that $\{B_1, \dots, B_n\} \subseteq I$ for every A .
- Then we conclude that $I \models_{strong} P$.

2 Question 2

This question was answered in class.

References

- [1] V.S. Subrahmanian. *Logic Programming*. Piazza. Accessed: 2014-10-04.